

Electroacupuncture Analgesia in Major Abdominal and Pelvic Surgery: A Randomised Study

P Poulain, E Pichard Léandri, A Laplanche, F Montange, J Bouzy, J Truffa-Bachi

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Summary

Proponents of acupuncture anaesthesia have tended to look on it as a replacement for conventional anaesthesia. This randomised, controlled trial involving 250 cancer patients undergoing abdominal or pelvic surgery, has aimed to assess the use of electroacupuncture as the sole analgesic, but within a standard anaesthetic. It was found that while there was little difference between the per-operative requirement for anaesthetic drugs, all the non acupuncture group required fentanyl, compared to only 5% of the electroacupuncture group ($P<0.001$). Time to spontaneous respiration ($P<0.02$) and extubation ($P<0.001$) immediately post-operatively was reduced in the acupuncture group.

Transcutaneous electrical nerve stimulation (TENS) was used for post-operative pain relief in the acupuncture group, while the control group received non-narcotic analgesics. There was no significant difference in the numbers of patients complaining of pain, but the requirement for additional analgesia was less when TENS was used ($P<0.01$). There was no difference in the period of post-operative ileus as measured by the return of normal bowel sounds. However the return to normal self caring was significantly improved in the acupuncture group ($P<0.02$). Both practically and economically this is probably the most important beneficial aspect of electroacupuncture demonstrated by this study.

Key words

Acupuncture anaesthesia, Post-operative analgesia, Randomised controlled trial, TENS.

Introduction

The analgesic efficacy of acupuncture is widely known. In the sixties doctors throughout China were taught to use acupuncture for anaesthesia (5), following the precepts dictated by the Great Proletarian Cultural Revolution, with a claimed effective rate of around 90% (1). The exchange of knowledge between Eastern and Western medicine resulted in doctors combining and adapting both approaches with a view to improving existing

methods (6,13). And more recently, research into the physiology of pain has gone some way to providing a possible explanation of the mechanisms involved in acupuncture analgesia. Electroacupuncture (EA) combined with non-narcotic anaesthesia is thought to provide good analgesia per-operatively. The aims of our study were:

- to assess whether EA could be used as the sole component of analgesia during major abdominal and pelvic surgery
- to show if patients who received less medication recovered more quickly
- to assess the analgesic efficacy of transcutaneous electrical nerve stimulation (TENS) during the 5 post-operative days.

Patients and methods

The day before surgery the patients were examined by the anaesthetist who explained the nature of the study and obtained informed consent. Patients were reassured that should EA prove to be ineffective, the standard method of analgesia would be adopted. The anaesthetist also taught all the patients how and when to use a TENS machine.

Inclusion criteria:

Cancer patients
Abdominal and/or pelvic surgery
Informed consent available

Exclusion criteria:

Infection or radiotherapy proximal to acupuncture points
Allergies (special premedication)
< 14 years old
Surgical emergencies
Pace makers
Anticipation of intensive post-operative therapy
Informed consent not available.

After informed consent had been obtained, patients were entered into the study and randomly allocated, using pre-prepared envelopes, into two groups. The control group received standard anaesthesia using narcotic analgesia per-operatively and non-narcotic analgesics post-operatively. Patients in the study group received EA during the operation and TENS

Table 1
METHOD OF ANALGESIA

	Control group	Study group
Per-operative	Narcotic	EA
Post-operative	Non-narcotic	TENS

post-operatively (Table 1).

In both groups patients received the same premedication 1 hour prior to surgery:

Flunitrazepam 3×10^{-2} mg kg⁻¹,
Atropine sulphate 0.25 to 0.75mg
(according to weight).

In Group 1, anaesthesia was induced using:

Droperidol 0.3mg kg⁻¹
Thiopentone 5mg kg⁻¹
Pancuronium bromide 8×10^{-2} mg kg⁻¹
Fentanyl 5×10^{-3} mg kg⁻¹

After intubation patients were ventilated with a 10ml kg⁻¹ mixture of oxygen and nitrous oxide (FiO₂ = 0.4) at a rate of 12 to 15 insufflations per minute.

In the Study group, 30 minutes prior to induction, 8 stainless steel, sterile acupuncture needles (diameter 0.4mm, length 40mm) were inserted to a depth of 5 to 10mm, bilaterally in the following 4 acupuncture points (Table 2): Ear *Shenmen*, PC.6, SP.6 and GB.39 (2).

Table 2
POINTS USED FOR ACUPUNCTURE ANALGESIA (2)

Shenmen ear point: inferior corner of bifurcation point of antihelix
Neiguan (PC.6): 2 acupuncture units of measurement (AUM or cun) above wrist, between tendons of palmaris longus and flexor carpi radialis
Sanyinjiao (SP.6): 3 AUM above tip of medial malleolus posterior to tibial border
Xuanzhong (GB.39): 3 AUM above lateral malleolus between posterior border of fibula and tendons of peroneus longus and brevis

These needles were connected to an electrostimulator (*DSM 1500*, MCM Laboratories, Paris) delivering a biphasic current at a frequency of 40Hz, raising the intensity until the subject reported a sensation of tingling (mean 5mA). After 30 minutes of electrostimulation, anaesthesia was induced in an identical manner to the Control group, using the same drugs at the same dosage, but without using *fentanyl*. Patients were also ventilated in the same way as those in the Control group. Immediately after induction of anaesthesia the electrostimulator current was increased to 15mA and the frequency was modulated.

During surgery, the anaesthetic requirements were assessed in both groups following the same clinical criteria: blood pressure and pulse rate were monitored every 5min. *Fentanyl*, in both groups, was injected to patients as required; per-operative assessment was made of each patient's dose requirements of anaesthetic drugs and of *fentanyl*.

At the end of the operation in the Study group patients, two sterile, conducting electrodes were placed either side of the scar and connected to a pre-set TENS machine (*Neuromod*, medtronic, USA). We made no pharmacological reversal of the competitive muscle relaxant (*pancuronium*), so patients were transferred to the recovery ward and extubated when clinical reversal, including spontaneous respiration with a minute volume of more than 100ml kg⁻¹, was evident.

Post-operatively, patients in both groups were given non narcotic analgesics as required, and were interviewed by a data manager over 5 days. Post-operative assessment was made for: quality of immediate recovery, dose requirements of analgesics, patient comfort and self assessment of pain, end of post-operative ileus, and achievement of independence.

A minimum of 125 subjects in each group was considered necessary to show a statistically significant decrease from 80% to 60% of pain experienced post-operatively, with a probability of 5%.

Results

During the period from November 1983 to December 1985, 250 cancer patients were entered into the study.

The data of 233 patients ($n=113$ Control group; $n=120$ Study group) were statistically analysed (12 patients from the pilot study were withdrawn: 2 whose randomised envelopes were lost, 2 for whom surgery was cancelled, and 1 who was randomised twice). The two groups were statistically matched as regards sex, age, weight, height and type of surgery

Table 3
PATIENT CHARACTERISTICS

	Control group (n=113)	Study group (n=120)
Female patients (%)	89%	86%
Average age	49	47
Average weight (kg)	62	61
Average height (cm)	164	162
Pelvic surgery (%)	88%	88%
Abdominal surgery (%)	12%	12%

Table 4
TYPE OF SURGERY

	Control group (n=113)	Study group (n=120)
Hysterectomy	32	24
Total hysterectomy	11	10
Total hysterectomy with pelvic & aortic lymphadenectomy	30	41
Nephrectomy	4	4
Cystectomy	3	5
Colectomy	2	4
Gastrectomy	1	1
Oophorectomy	6	4
Laparotomies	17	20
Pelvic and aortic lymphadenectomy	7	7

(Tables 3 and 4). There was no difference between the two groups in the anaesthetic dose requirements (related to body weight and duration of surgery) (Table 5). All the patients in the Control group required *fentanyl*, whereas only 7 out of the 120 patients in the Study group needed it, and that at very low dosages.

Table 5
PER-OPERATIVE DRUG REQUIREMENTS

	Control group (n=113)	Study group (n=120)	P
Duration of surgery (min)	146	152	NS
Bleeding (l)	0.7	0.9	NS
Average dose requirements ($\times 10^{-3}$ mg kg ⁻¹ min ⁻¹)			
Barbiturate (<i>thiopentone</i>)	61	69	NS
Neuroleptic (<i>droperidol</i>)	2.7	3.1	NS
Muscle relaxant (<i>pancuronium</i>)	1	1	NS
Patients requiring <i>fentanyl</i> (%)	100%	5%	<0.001

NS = Non significant

The anaesthetics were given by 11 anaesthetists, one of whom anaesthetised 53% of the patients. There was no significant difference between his anaesthetic management (of either group) and that of the others.

Peri-operatively patients in the Study group recovered spontaneous respiration ($P<0.02$) and were extubated ($P<0.001$) more quickly (Table 6). Patients in the Study group recovered spontaneous respiration a mean of 19min earlier than Control patients (74.1% of patients in Study and 55% in Control had recovered by 30min after surgery). Patients in the Study group were extubated at a mean of 36min earlier than those in the Control group (57.5% of patients in Study and 27.4% in Control were extubated by 60min after surgery).

Post-operatively, 226 patients out of the 233 had a complete 5 days follow-up; of the remainder: 2 deceased (Control and Study), 3 were returned to the recovery ward (2 Study, 1 Control), 1 was discharged on day 4 (Study) and 1 escaped on day 3 (Study). The percentages of patients in the Study

Table 6
IMMEDIATE POST-OPERATIVE RECOVERY (RELATED TO SURGERY TIME)

	Control group (n=113)	Study group (n=120)	P
Time to spontaneous respiration (min per min of surgery)	0.38	0.21	<0.02
Extubation time (min per min)	0.64	0.32	<0.001

Table 7
POST-OPERATIVE BENEFIT

	Control group (n=113)	Study group (n=120)	P
Patients complaining of pain	81%	75%	NS
Requiring analgesics	26%	18%	<0.01
Self-caring within 3 days	20%	32%	<0.02

NS = Non significant

group using the TENS machine from day 1 to day 5 were respectively 81%, 79%, 70%, 54%, 18%.

There was no statistical difference between the two groups in those patients who complained of pain; conversely, patients in the Study group required less analgesia ($P<0.01$) (Table 7).

No difference was observed in the recovery time from post-operative ileus: the Control group had a mean of 3.2 days, the Study group a mean of 3.3, but patients in the Study group became self-caring sooner ($P<0.02$); during the first 3 days 20% of patients in the Control group and 32% in the Study group had achieved their independence.

Discussion

Most of the works discussing the clinical analgesic efficacy of EA for surgery are descriptive, and therefore of limited value (1,3,4,6,10,11). Of the recent studies which have been carried out, few have been randomised (3,8,15) and the results are unconvincing. This trial was randomised and showed a statistically significant difference between the two groups in dose requirement of opiates, while patients required the same dose of non-narcotic anaesthetic drugs.

One can argue that this study was not blind and that this might have affected the responses of both patients and physicians. However, this work involved 11 different anaesthetists, most of whom were not accustomed to acupuncture and who did not really believe that this method of anaesthetising patients was efficient. Comparison between anaesthetists showed no difference in the dose of drugs administered. It is, anyway, very difficult to use a double blind method in the assessment of the effect of EA. In our protocol we could have used sham acupuncture versus standard acupuncture, but previous studies have shown that sham and standard acupuncture were effective in about 50% and 65% respectively (9). Thus, the number of patients in the trial needed to achieve statistical significance would have been substantially larger.

The combination of Chinese and Western medicine has encouraged more research, resulting in a better understanding of the possible mechanisms underlying EA (7,14). Physicians differ in their opinion as to the role of EA during surgery, but there are two main schools of thought. Most Eastern doctors and some Europeans try to replace opiate and anaesthesia with acupuncture alone (12). On the other hand, in our trial, we proposed that acupuncture induced analgesia was just a component of anaesthesia, and so we combined the two techniques, for the comfort of the patient and of the medical team. The electrostimulation of selected, traditional acupuncture points enabled us to give an effective analgesia. The low intensity currents, with modulated frequency are thought both to stimulate the natural secretion of endorphins (low frequency) and serotonin (high frequency), and to provide segmental analgesia by activation of the spinal gate control system.

The choice of points used in this trial was specific to the type of surgery (lower abdominal and pelvic) and may need to be adapted for other operations.

None of the drugs we used for anaesthesia is known to possess analgesic properties. In both groups, the concentration of N₂O provided a potentiation of the other drugs, with a negligible analgesic effect during major surgery.

EA during the surgical period provided a good analgesia and allowed us to extubate patients in the Study group in half the time required for those in the Control group, suggesting that EA affords a safer immediate post-operative period.

In both groups there were patients who complained of post-operative pain, but those in the Study group required less analgesia. At first sight this seems a contradiction, however we think the reduced use of opiate analgesia allowed EA patients to be more alert and aware, and thus to recall any pain experienced more clearly.

We were expecting that EA patients would recover their bowel sounds more quickly than Control group patients, because opiates are known to inhibit peristalsis. In fact, there was no statistical difference in recovery between the groups. It may be that post-operative ileus is a local effect related to the surgery, but the endogenous opioids released following EA certainly act on opiate receptors in the gut, and constipation has been recorded as a complication of acupuncture (16). However, we have no explanation of why patients in the Study group recovered and became self caring more quickly following surgery than controls, but both practically and economically, it is probably the most important beneficial aspect of EA in this trial.

Conclusion

The results of this trial suggest that in major abdominal and pelvic surgery, EA is an effective component of anaesthesia. It is a safe technique, providing a shorter post-operative recovery time, with equally effective pain control to those patients receiving opioid analgesia.

We therefore recommend this combination of electroacupuncture with conventional, non-opiate anaesthesia, particularly in the elderly and those with respiratory insufficiency.

**P Poulain MD, E Pichard Léandri MD
F Montange MD, J Truffa-Bachi MD**
Service d'Anesthésie, Institut Gustave-Roussy

A Laplanche MD, J Bouzy
Département de Biostatistiques et d'Epidémiologie
Institut Gustave-Roussy
Rue Camille Desmoulins
94805 Villejuif Cedex, France

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